Formula 1 Race Intelligence Dashboard – Project Report

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1. Introduction

1.1 Project Overview

The Formula 1 Race Intelligence Dashboard is a Tableau project that turns 6,464 race records from 17 seasons (2010 to 2026) into one interactive view of driver performance, team championship trends, tyre strategy, circuit participation, driver reliability and podium results. It was built as part of the Data Analytics with Tableau program on SkillWallet.

The project covers the full analytics cycle: data collection from a multi-sheet dataset, data preparation, nine visualizations, a combined dashboard, a data story, performance testing, publishing to Tableau Public, and embedding in a Flask web application. Team strategists, race engineers, motorsport journalists and F1 fans can use it to find patterns, spot strengths and weaknesses, and make data-driven decisions.

1.2 Purpose

The purpose is to give stakeholders a single place to explore Formula 1 data visually instead of reading raw tables. The project is built around three user scenarios:

- **Rajan, F1 Team Performance Strategist.** He works at a mid-tier constructor and needs to benchmark his team's championship points against Red Bull Racing and Mercedes, see how tyre compounds (Soft, Medium, Hard, Intermediate) affect results, and find which drivers and nationalities form the strongest talent pools.
- **Divya, Motorsport Broadcast Analyst.** She needs a Driver Performance Leaderboard, the countries that hosted the most Grands Prix, and podium distributions by driver and team, for on-screen graphics and editorial pieces.
- **Arvind, FIA Technical Regulation Analyst.** He studies DNF (Did Not Finish) patterns (engine failure, brake failure, collision, mechanical issues) to see whether failures cluster in particular teams or circuit types, and whether regulation changes improved race completion rates.

2. Ideation Phase

2.1 Problem Statement

F1 race data is large and spread across several tables (race results, qualifying, lap intelligence, circuits, championship standings), so teams, broadcasters and regulators struggle to answer simple questions quickly: which teams and drivers perform best, how reliable are they, and what strategy choices drive results. This project builds a filterable Tableau dashboard that answers those questions for the three user types below.

Who	Needs to	Because
Team strategist (Rajan)	Benchmark points trajectory and tyre strategy against top teams	To plan race and recruitment strategy and climb the Constructors' standings
Broadcast analyst (Divya)	Pull leaderboards, hosting countries and podium counts fast	To produce pre-race and post-race content and infographics
Regulation analyst (Arvind)	Find where and why DNFs concentrate	To back technical regulation recommendations with evidence

2.2 Empathy Map Canvas

Built for Rajan, the primary user.

Quadrant	Content
Says	"How do we compare with Red Bull and Mercedes?" "Which tyre strategy works at which circuit?"
Thinks	Data is scattered and slow to compare; the right talent and strategy could move us up the standings
Does	Reviews race results, championship points and driver stats by hand across multiple sheets
Feels	Pressure to deliver results; frustrated by manual analysis and unclear trends
Pains	Raw data is hard to read; no single view of drivers, teams, tyres and reliability

Quadrant	Content
Gains	One interactive dashboard with filters that shows strengths, weak spots and clear next steps

2.3 Brainstorming

Ideas considered for the dashboard, and what was kept:

- KPI cards for the headline numbers (teams, seasons, race outcomes, circuits): **kept**
- Geographic map of driver nationalities across Grand Prix nations: **kept**
- Stacked bar of DNF reasons per driver for reliability: **kept**
- Leaderboard with a Top N parameter so users choose how many drivers to see: **kept**
- Team ranking and team championship comparison charts: **kept**
- Tyre compound usage as a pie chart: **kept**
- Country-wise participation treemap and podium packed bubble chart: **kept**
- Heavy calculated fields and predictive models: **left out**, since the source data already had the needed metrics (see Future Scope)

3. Requirement Analysis

3.1 Customer Journey Map

The journey of a typical user (Rajan, the team strategist) from question to decision:

Stage	What the user does	Touchpoint	Experience
Awareness	Needs to benchmark the team before the season	Shared Tableau Public link or Flask web page	Curious, wants quick answers
Explore	Reads the KPI cards and global driver map	Dashboard top section	Gets the big picture at once
Analyze	Filters by Year and DNF Reason; sets Top N	Filters and parameter	Finds team, driver and tyre patterns
Decide	Compares team championship bars and podium bubbles	Team and podium charts	Confident in the strategy and recruitment direction
Share	Sends the story or embeds it in a report	Story and Flask embed	Easy to present to others

3.2 Solution Requirement

Functional requirements

- Load the multi-sheet F1 dataset into Tableau and join the sheets
- Clean missing values, duplicates and inconsistencies, and standardize formats
- Provide nine visualizations covering KPIs, map, reliability, leaderboard, team ranking, championship, tyres, country participation and podiums
- Provide filters for DNF Reason and Year of Race Date, and a Top N parameter (default 10)
- Combine the views into one dashboard and a guided data story
- Publish to Tableau Public and embed in a Flask web application

Non-functional requirements

- All visualizations respond within acceptable time across every filter combination
- The dashboard is readable by non-technical viewers and needs no Tableau license to view
- Charts accurately reflect the source race data

3.3 Data Flow Diagram

1. The F1 dataset (5 sheets, about 13,750 rows) is downloaded from the source database.
2. Tableau Desktop connects to it and joins the sheets, with race_results (6,464 rows, 41 columns) as the main table, using an extract.

3. Data preparation fixes missing values, duplicates and formats.
4. Worksheets (9 visualizations) are built on the prepared data.
5. Worksheets combine into the dashboard and then the story.
6. The workbook is published to Tableau Public.
7. The embed code is placed in an HTML template served by Flask, which viewers open in a browser.

3.4 Technology Stack

Layer	Tool
Data source	Multi-sheet F1 dataset (race results, qualifying, lap intelligence, circuits, championship standings)
Data preparation and visualization	Tableau Desktop (extract mode)
Publishing	Tableau Public
Web integration	Flask (Python), HTML templates, Tableau Public embed code
Learning platform	SkillWallet (Data Analytics with Tableau program)

4. Project Design

4.1 Problem Solution Fit

Problem	Solution feature
Data is spread over several sheets and hard to compare	One joined, cleaned data source feeding a single dashboard
Teams cannot quickly benchmark against Red Bull and Mercedes	Team Performance Ranking and Team Championship Performance Analysis
Strategists need tyre and talent insight	Tyre Compound Usage pie, Driver Performance Leaderboard, Global Driver Distribution map
Regulators need evidence on failures	Driver Reliability Analysis stacked by DNF reason, filterable by year
Broadcasters need ready visuals	Podium bubble chart, country participation treemap, Top N leaderboard

Problem	Solution feature
Results must be easy to share	Tableau Public publishing, a data story and a Flask web embed

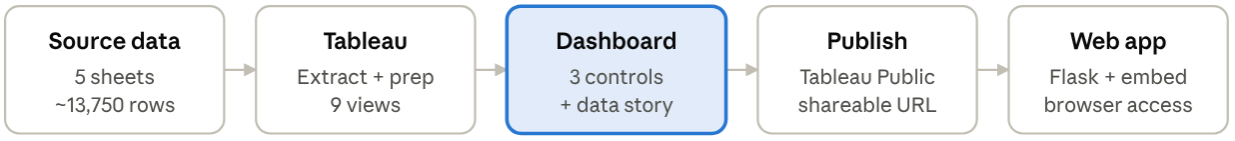
4.2 Proposed Solution

A nine-visualization Tableau dashboard titled "Formula 1 Race Intelligence Dashboard", with three controls (DNF Reason filter, Year of Race Date filter, Top N parameter), supported by a story named "Formula 1 Race Intelligence: From Global Overview to Championship Success". Both are published to Tableau Public and embedded in a Flask website so anyone can explore them in a browser without a Tableau license.

4.3 Solution Architecture

Data moves from the source sheets through Tableau to the web, as shown below.

F1 data flows from source sheets through Tableau to a Flask web app



solution architecture · 5 stages

Viewers reach the dashboard through the Flask page or the Tableau Public link, and filters work the same in both.

5. Project Planning & Scheduling

5.1 Project Planning

This is an individual project, assigned to Hardik Sachdeva on SkillWallet and split into eight epics. Five of them are rated Medium complexity with a 5 hour duration each.

#	Epic	Key tasks	Complexity	Duration
1	Data Collection & Extraction from Database	Download the dataset; connect it with Tableau	Medium	5h

#	Epic	Key tasks	Complexity	Duration
2	Data Preparation	Clean missing values, duplicates and inconsistencies; standardize formats	Medium	5h
3	Data Visualization	Build the 9 unique visualizations	Medium	5h
4	Dashboard	Combine views into one responsive, interactive dashboard	Medium	5h
5	Story	Build the narrative with multiple scenes	Medium	5h
6	Performance Testing	Data volume, filter use, calculation fields, number of visualizations	(fill in)	(fill in)
7	Web Integration	Publish to Tableau Public; embed in a Flask app	(fill in)	(fill in)
8	Project Demonstration & Documentation	Demo link, GitHub link, this report	(fill in)	(fill in)

6. Functional and Performance Testing

6.1 Performance Testing

Performance testing checked how the volume of data, the filters and the number of visualizations affect how quickly the dashboard responds.

Factor	Finding
Amount of data rendered to Tableau	The dataset has 5 sheets and about 13,750 rows. The main race_results sheet has 6,464 rows and 41 columns. Tableau handled it efficiently in extract mode, and all 9 visualizations loaded within acceptable time across every combination of DNF Reason, Year and Top N.

Factor	Finding
Utilization of filters	Three controls: a DNF Reason dropdown (Engine failure, Brake failure, Collision, Mechanical, Other) that cross-filters the Driver Reliability chart and race results; a Year of Race Date dropdown for season-wise or multi-season views from 2010 to 2026; and a Top N integer parameter (default 10) that sets how many drivers show on the Driver Performance Leaderboard.
Number of calculation fields	Minimal: only Top N filtering and simple parameters. Race results, championship points, lap times, tyre compounds, weather, podium finishes, circuits, teams and drivers were already in the data, so no complex calculated fields were needed. This keeps the dashboard fast and faithful to the original data.
Number of visualizations / graphs	9 visualizations: 4 KPI cards, a geographic map, a stacked bar, a vertical bar, a horizontal bar, a grouped vertical bar, a pie chart, a treemap or highlight table, and a packed bubble chart.

7. Results

7.1 Output Screenshots

The final dashboard, "Formula 1 Race Intelligence Dashboard", brings all nine visualizations into one interactive view with DNF Reason, Year of Race Date and Top N controls. Add a clean Tableau screenshot under each row below.

#	Visualization	Chart type	What it shows	Screenshot
1	KPI cards	4 cards	10 competing teams, 17 seasons in play, 6,464 race outcomes logged, 29 circuits showcased	[insert]
2	Global Driver Distribution Across Grand Prix Nations	Geographic map	Where drivers come from and where races are held	[insert]

#	Visualization	Chart type	What it shows	Screenshot
3	Driver Reliability Analysis	Stacked bar, by DNF reason	Which drivers lose races to engine, brake, collision, mechanical or other failures	[insert]
4	Driver Performance Leaderboard	Vertical bar, Top N controlled	The top N drivers (default 10)	[insert]
5	Team Performance Ranking	Horizontal bar	Teams ranked by performance	[insert]
6	Team Championship Performance Analysis	Grouped vertical bar	Championship points per team, Red Bull Racing and Mercedes highest, then Ferrari	[insert]
7	Tyre Compound Usage Distribution	Pie chart	Share of Soft, Medium, Hard and other compounds	[insert]
8	Country-wise Race Participation	Treemap / highlight table	Participation by country	[insert]
9	Podium Achievements by Driver and Team	Packed bubble chart	Podium finishes by driver and team	[insert]

Story. The story, "Formula 1 Race Intelligence: From Global Overview to Championship Success", walks the reader from the global map through to championship results in scenes.

Publishing. The dashboard and story were published to Tableau Public via File > Save to Tableau Public, giving a shareable URL with full filtering for any viewer, no license needed.

Web integration. The Tableau Public embed code sits in an HTML template rendered by a Flask route and served from a local development server. The Race Intelligence site has a landing page, headline statistics, an "Intelligence Dimensions" section, and the embedded dashboard and story, all keeping the published filters and parameters.

8. Advantages & Disadvantages

Advantages

- Brings nine views of 6,464 race records into one interactive dashboard
- Filters and the Top N parameter let each user explore their own question

- Fast, because it uses an extract and very few calculated fields
- Anyone can view it through a link or the Flask site, with no Tableau license
- Serves strategists, broadcasters and regulators with the same data

Disadvantages

- Data is a fixed extract, not a live feed, so it must be refreshed by hand
- The Flask app runs on a local development server and is not yet deployed publicly
- No predictive models; the dashboard describes the past
- Tableau Public workbooks are public and have data size limits

9. Conclusion

The project turned a multi-sheet Formula 1 dataset into a fast, interactive Tableau dashboard and story that answer the questions of team strategists, broadcast analysts and regulation analysts. It covers team and driver performance, tyre strategy, reliability and podium results across 17 seasons, and it is shared through Tableau Public and a Flask web page. Keeping calculations minimal let the dashboard stay quick and true to the source data.

10. Future Scope

- Predictive models for championship outcomes and podium probability
- Live or scheduled data refresh so new races appear automatically
- Lap-by-lap and pit-stop strategy views using the lap intelligence data
- Weather and circuit-type filters
- Deploying the Flask site to a public host with user accounts and saved views
- Mobile-first dashboard layouts

11. Appendix

Item	Details
Source code	Flask app (app.py) and HTML template with the Tableau embed code: (paste here)
Dataset link	F1 Racing Project Dataset: (paste link here)
GitHub link	(paste link here)
Project demo link	(paste Tableau Public / demo link here)